P07

Epidemic algorithms for replicated database maintenance

Authors	Alan J. Demers, Dan Greene, Carl H. Hauser, Wes Irish, John Larson
Published	ACM Symposium on Principles of Distributed Computing (PODC) (1987)
DOI	https://doi.org/10.1145/41840.41841
Paper Link	https://dl.acm.org/doi/pdf/10.1145/41840.41841
Domain	Networking & distributed systems

Abstract

When a database is replicated at many sites, maintaining mutual consistency among the sites in the face of updates is a significant problem. This paper describes several randomized algorithms for distributing updates and driving the replicas toward consistency. The algorithms are very simple and require few guarantees from the underlying communication system, yet they ensure that the effect of every update is eventually reflected in all replicas. The cost and performance of the algorithms are tuned by choosing appropriate distributions in the randomization step. The algorithms are closely analogous to epidemics, and the epidemiology literature aids in understanding their behavior. One of the algorithms has been implemented in the Clearinghouse servers of the Xerox Corporate Internet, solving long-standing problems of high traffic and database inconsistency.

Real-World Impact & Application

Amazon DynamoDB, Apache Cassandra & Blockchain: Demers et al. invented epidemic/gossip algorithms to maintain distributed databases without central locks or single points of failure. Today, gossip protocols are the backbone of hyper-resilient distributed systems worldwide, including Amazon DynamoDB, Apache Cassandra, and the peer-to-peer transaction broadcasting networks of Bitcoin and Ethereum.